INCH-POUND
MS18196F
w/AMENDMENT 2
28 October 2013
SUPERSEDING
MS18196F
w/AMENDMENT 1
3 June 2013

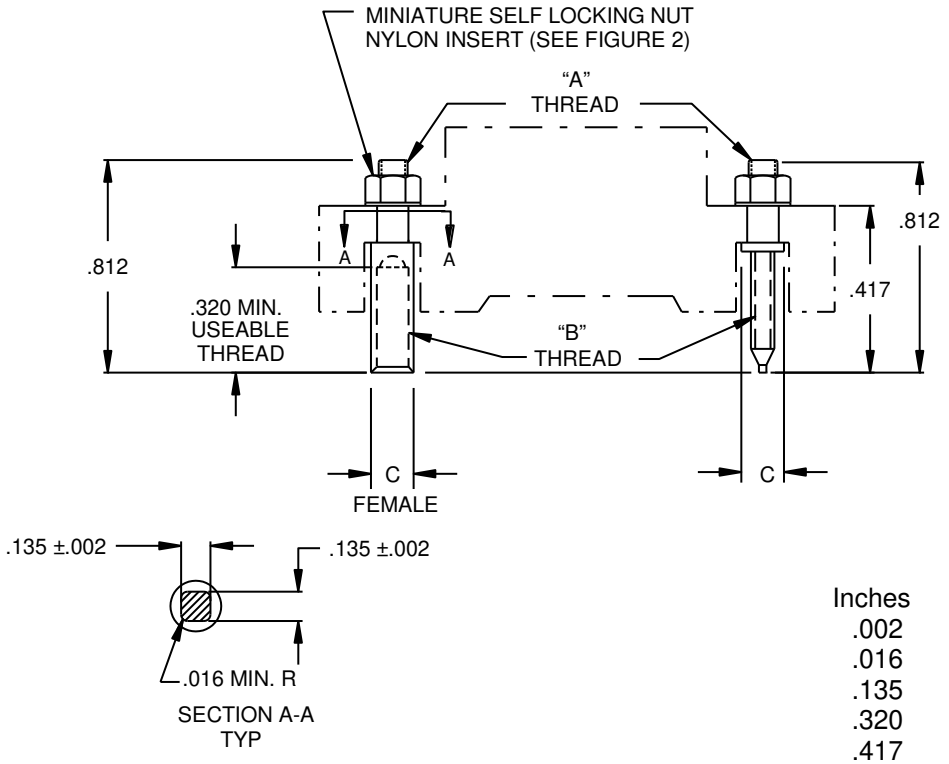
DETAIL SPECIFICATION SHEET

JACKSCREW, ELECTRICAL CONNECTOR, MALE AND FEMALE, FIXED

Reinstated after 10 July 2009 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28748.



Inches	mm
.002	0.05
.016	0.41
.135	3.43
.320	8.13
.417	10.59
.812	20.62

FIGURE 1. Fixed jackscrew male and female.

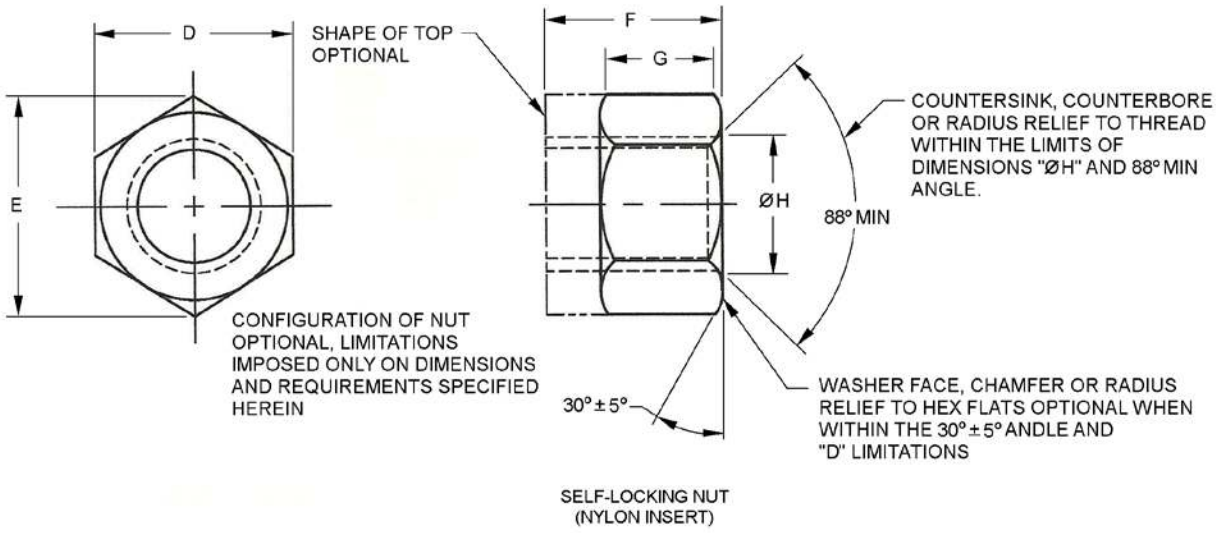
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Fixed jackscrew designator	Gender	Connector size (number of contacts)	A Thread class 2	B Thread class 2	C dia max
MS18196-1	Male	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	6-32 UNC-2A .138-32 UNC	.190 (4.83)
MS18196-2	Female	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	6-32 UNC-2B .138-32 UNC	.190 (4.83)
MS18196-3	Male	104	6-32 UNC-2A .138-32 UNC	8-32 UNC-2A .164-32 UNC	.137 (3.48)
MS18196-4	Female	104	6-32 UNC-2A .138-32 UNC	8-32 UNC-2B .164-32 UNC	.137 (3.48)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material corrosion resistance steel in accordance with ASTM A276, ASTM A473, ASTM A581/A581M, ASTM A582/A582M, SAE AMS5636, SAE AMS5639, SAE AMS5648, classes 302, 303, 303Se, 304 or 316 except for aluminum alloy knob and .040 inch (1.02 mm) flat washer.
5. Finish passivated in accordance with ASTM A967 or SAE AMS2700 except for anodized knob.
6. Jackscrew may have .005 inch (0.13) minimum vertical float.

FIGURE 1. Fixed jackscrew male and female – Continued.

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Self-locking nut dimensions

Self-locking nut Part or Identify Number (PIN) (See note 9)	Connector size (number of contacts)	Thread AS8879	D		E ref	F		G ref	H
			min	max		min	max		
MS21044-C04	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	.1120-40 UNJC-3B	.243 (6.17)	.251 (6.38)	.268 (6.81)	.133 (3.38)	.157 (3.99)	.044 (1.12)	.136 (3.45)
MS21044-C06	104	.1380-32 UNJC-3B	.305 (7.75)	.313 (7.95)	.339 (8.61)	.160 (4.06)	.188 (4.78)	.072 (1.83)	.161 (4.09)

NOTES:
1. Self-locking nut shown for reference purposes only.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Unless otherwise specified, tolerances are ±.016 inch (0.41mm).
5. Material corrosion resistance steel in accordance with ASTM A276, ASTM A473, ASTM A581/A581M, ASTM A582/A582M, SAE AMS5636, SAE AMS5639, SAE AMS5648, classes 302, 303, 303Se, 304 or 316.
6. Finish passivated in accordance with ASTM A967 or SAE AMS2700.
7. Not to be used where temperatures exceed 250°F (121°C).
8. Do not reuse nylon nut after removal.
9. National Aerospace Standard NASM21044. Not to be ordered with this slash sheet.
10. Copies of NASM21044 are available online at http://www.aia-aerospace.org/, or may be purchased from Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3901.

FIGURE 2. Self-locking nut.

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Fixed jackscrew male and female. See figure 1.
Self-locking nut. See figure 2.

Qualification: Qualified sources for MIL-DTL-28748/1 through MIL-DTL-28748/6 that meet the qualification testing requirements in MIL-DTL-28748 require no further testing for the hardware.

Qualification of hardware only: the hardware shall be installed in a qualified connector and shall be subjected to the tests below in accordance with MIL-DTL-28748. Sample size, if qualifying all hardware configurations, shall be two connector pairs.

Visual and mechanical inspection
Vibration
Shock
Salt spray

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-28748, this document references the following:

MIL-DTL-28748/1
MIL-DTL-28748/2
MIL-DTL-28748/3
MIL-DTL-28748/4
MIL-DTL-28748/5
MIL-DTL-28748/6
ASTM A276
ASTM A473
ASTM A581/A581M
ASTM A582/A582M
ASTM A967
NASM21044
SAE AMS2700
SAE AMS5636
SAE AMS5639
SAE AMS5648
SAE AS8879

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army - AM, AR, AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 99

(Project 5935-2013-212)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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